ENSF 380

TOPIC 26: DATABASES PART II

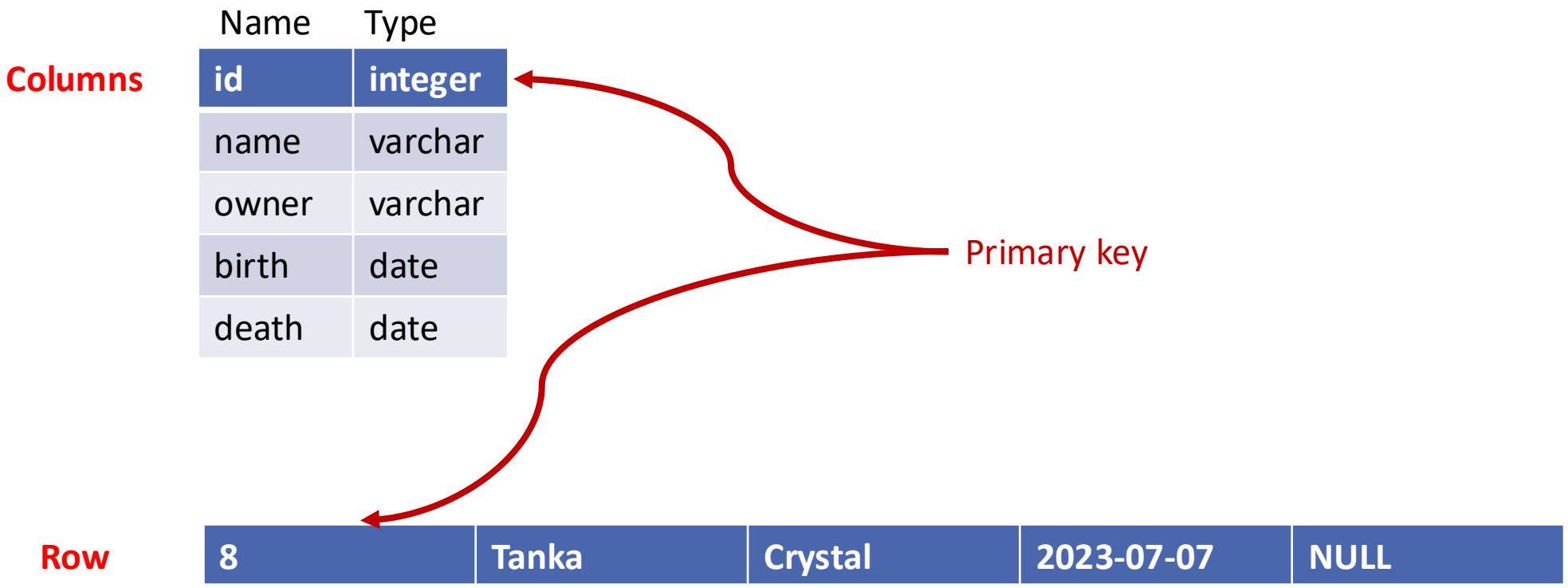
Reminders

Relational databases represent data using tables (relations), rows (tuples), and columns (attributes)

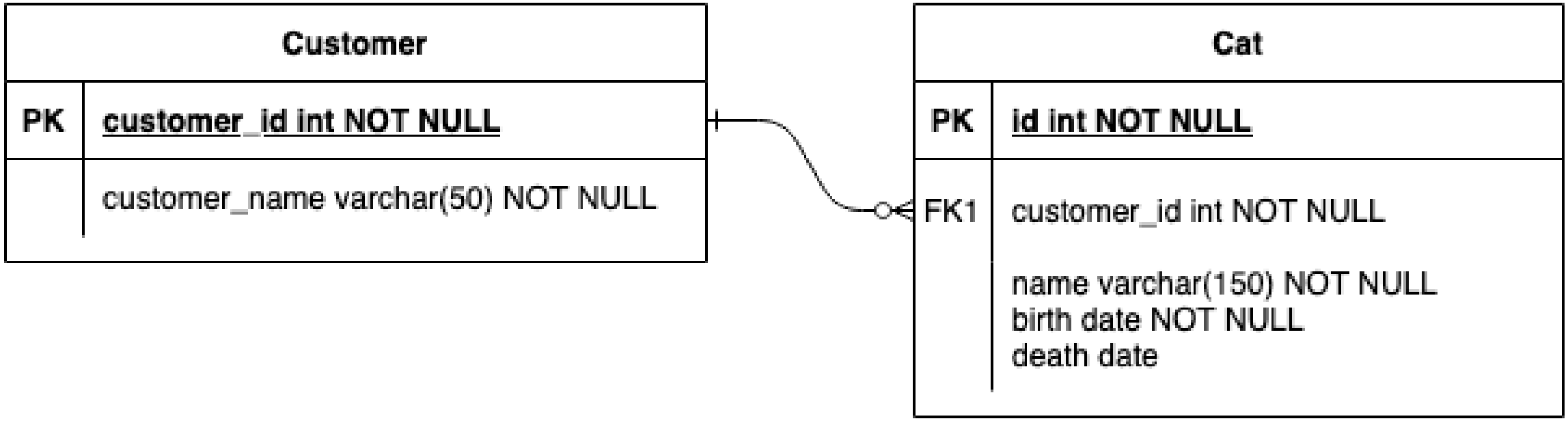
SQL is a language used to manage and retrieve data in relational databases

Applications interact with a database by generating queries and transactions

Relational Databases



Relational Databases



Relations Versus Objects

Relational Databases

Customer

id	firstname	surname
2	Karl	Kaufmann

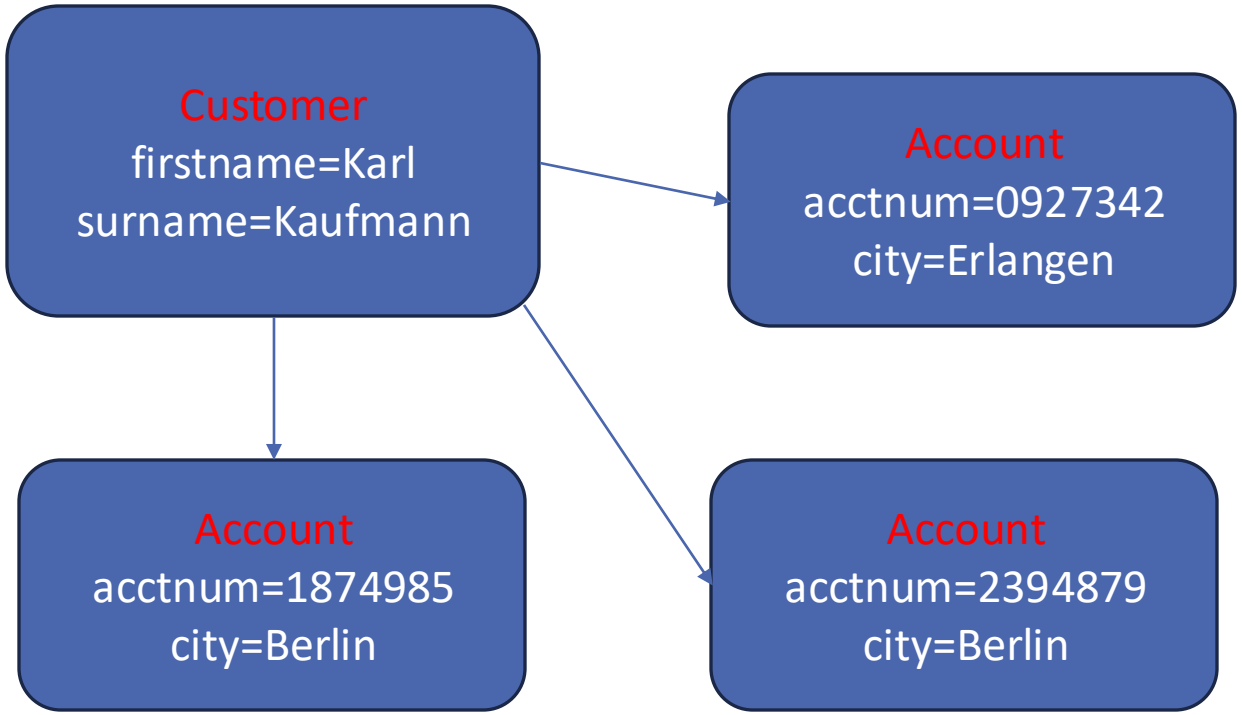
Branch

id	city
6	Berlin
9	Erlangen

Account

id	acctnum	branch	customer
1	1874985	6	2
2	2394879	6	2
3	0927342	9	2

Object-Oriented Design



JDBC – Java Database Connectivity

Included in JDK

Application programming interface (API) for connecting to databases

Requires a driver for the intended database

- Download the appropriate driver
- Add .jar to classpath

Steps for Connecting to a Database with Java

Step 1: Create a connection

Step 2: Create a statement

Step 3: Execute the query

Step 4: Process the retrieved results

Step 5: Close the connection and release resources

Need multiple try/catch statements with SQLExceptions in case any stage fails

Step 1: Create a connection

Previous versions required driver registration

- From JDBC 4.0 and onwards, drivers in the classpath are automatically loaded

Three necessary Strings:

- Database url: location and name of the database
- Username: designated database user (root, etc.)
- Password: corresponding account password

Example:

```
Connection myConnect = DriverManager.getConnection("jdbc:postgresql://localhost/pets", "oop", “ucalgary");
```

Step 2: Create a statement

Three statement types

- `Statement`: Simple SQL statement with no parameters
- `PreparedStatement`: SQL statements with input parameters
- `CallableStatement`: used for stored procedures with input/output parameters

Statement:

```
Statement myStmt = dbConnect.createStatement();
```

Prepared Statement:

```
String query = "INSERT INTO cat (name, owner, birth) VALUES (?,?,?)";
PreparedStatement myStmt = dbConnect.prepareStatement(query);

myStmt.setString(1, name);
myStmt.setString(2, owner);
myStmt.setDate(3, birthdate);
```

Step 3: Execute the statement

Three execute options

- `execute`: returns true if first object returned by the query is a `ResultSet` object
- `executeQuery`: returns one `ResultSet` object
- `executeUpdate`: returns an integer value for the number of rows affected by the statement (needed for transactions that update, insert, or delete)

Statement:

```
ResultSet results = myStmt.executeQuery("SELECT * FROM cat");
```

Prepared Statement:

```
int rowCount = myStmt.executeUpdate(); //after preparing an insert
```

Step 4: Process the retrieved results

01_DemoApplication

A `ResultSet` object stores the results of a query

Allows the data results to be used within an application

https://docs.oracle.com/javase/tutorial/jdbc/basics/retrieving.html

```
try{
   Statement myStmt = dbConnect.createStatement();
   results = myStmt.executeQuery("SELECT * FROM cat");

   while (results.next()){
     System.out.println("Print results: " + results.getString("name") + ", " +
     results.getString("owner"));

     catsAndOwners.append(results.getString("name") + ", " + results.getString("owner"));
     catsAndOwners.append('\n');
   }
 } catch (SQLException ex) {
     ex.printStackTrace();
 }
```

Step 5: Close the connection and release resources

Good practice to release resources

Close all statement objects after use

Close `ResultSet` objects

Close `Connection` objects

Use built-in `close()` methods

Exercise 26.1

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Exercise 26.2

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Summary: Knowledge

Java Database Connectivity (JDBC)

Retrieval of information from a database using Java

Modifying a database using Java

Relational databases versus object-oriented programming

Summary: Skills

Using JDBC to access a PostgreSQL database

Creating a database connection

Creating and executing statements

Creating and executing prepared statements with input parameters

Using retrieved data

Releasing resources upon completion

Additional Information

Related reading

- Volume 2, Chapter 5: Database Programming